

Praneel Magapu

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Education

North Carolina State University

Aug 2023 – May 2027

Bachelor of Science in Computer Science (GPA: 3.5 / 4.0)

Raleigh, NC

- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Software Engineering, Artificial Intelligence, Machine Learning; two independent research studies

Experience

Align Technology

May – Aug 2026

Software Engineering Intern, Product R&D

Raleigh, NC

- Redesigned support-structure generation in a large production C++ pipeline for directly 3D-printed medical devices, improving surface quality while reducing downstream manual finishing and material use.
- Implemented mesh-processing algorithms with Eigen and Open3D that derive valid placement regions from part geometry, so the system generalizes across device shapes instead of requiring per-case operator tuning.
- Applied OpenCV image processing to rasterized geometry to compute pixel-level contact footprints, tuning the interface between generated structures and the part surface for clean detachment after printing.

DevDynamics.ai

May – Aug 2025

Software Engineering Intern

Remote

- Built the backend intelligence layer for InboxFlow, a Gmail workflow service that classifies transactional email and tracks order, return, and refund state across a thread.
- Designed a two-stage classifier that runs deterministic regex on the fast path and escalates only low-confidence messages to a Gemini LLM, mapping structured model output back into a fixed label schema.
- Migrated thread-timeline persistence from memory to Supabase (PostgreSQL) and deployed the Flask service to AWS Elastic Beanstalk as a stable public endpoint for a Gmail Add-on.

NC State Center for Geospatial Analytics

Aug 2026 – Present

Undergraduate Research Assistant, GAEC Lab

Raleigh, NC

- Extend a harmonized flood-segmentation training pipeline by integrating five published benchmark corpora, reconciling conflicting coordinate reference systems, resolutions, sensor modalities, and label taxonomies.
- Contribute to FloodAtlas, fusing Sentinel-1 SAR with optical imagery to detect ephemeral flooding at global scale.

Liquid Rocketry Lab, NC State

Sep 2024 – May 2025

Data Engineer

Raleigh, NC

- Built the real-time telemetry pipeline for live rocket engine hot fire tests in Python and MongoDB across containerized microservices, capturing abort and breach events at sub-second latency.
- Designed an observable fault-detection pattern that decouples threshold logic from the ingestion path, so new fault conditions ship by subscribing a detector rather than by editing the pipeline.

Projects

Mendacia | Python, Flask, React, TypeScript, TwelveLabs, LLM APIs

Hack_NCState 2026

- Built a full-stack multimodal media forensics platform that turns raw video into a structured, explainable report flagging propaganda techniques and cross-modal inconsistencies.
- Designed a defensive adapter layer that normalizes backend responses into a strict UI schema before they reach application state.

Mesh Alignment Engine | Python, Open3D, NumPy, trimesh

May 2026

- Implemented ICP surface mesh registration from first principles, estimating rigid transforms through SVD of the cross-covariance matrix, with a coarse-to-fine strategy to speed convergence.
- Built a per-point deviation heatmap so alignment quality is verifiable visually, not just numerically.

Technical Skills

Languages: C++ (C++17), Python, Java, C, SQL, JavaScript, TypeScript, Bash

Frameworks: Flask, FastAPI, Pydantic, React, Tailwind CSS, Eigen, Open3D, OpenCV, NumPy, trimesh

Infrastructure: AWS (Elastic Beanstalk), Docker, GitHub Actions, Git, CMake, PostgreSQL, Supabase, MongoDB

Concepts: LLM integration, RAG systems, multimodal AI, semantic segmentation, REST API design, CI/CD